



Assignment Part 1: Data Collection, Understanding, and Visualization for Risk-Return
Analysis and Portfolio Optimization

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DAMO611 – Data Analytics Case Study 3

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1. Data Collection

Statement of Purpose

This project tackles a common problem for investors: how to wisely spread their money across different parts of the market. Often, traditional investment methods don't fully consider how different assets relate to each other, the specific challenges of each industry, or the real benefits of diversification. This can lead to less-than-ideal results.

Our goal is to address this issue by utilizing historical stock data to understand the risk and return patterns in the Finance, Technology, and Energy sectors. By analyzing these insights, we will identify and build the best possible sector allocations, which will help create portfolios that maximize returns while keeping risk in check, all within practical, real-world limits.

Scope of the Project

This report focuses on the foundational phases of the project: Data Collection, Data Understanding, and Data Visualization. These initial stages are crucial for establishing a robust analytical framework upon which subsequent model building and evaluation will rely. Specifically, this document details the acquisition of financial data, its initial exploration to uncover key metrics and characteristics, and the creation of visual representations to effectively communicate observed patterns, trends, and correlations. This comprehensive preliminary analysis lays the essential groundwork for addressing the project's core research questions and testing its formulated hypotheses.

Background Research and Literature

The theoretical foundation of this project is built upon Modern Portfolio Theory (MPT), a framework pioneered by Harry Markowitz (1952). MPT revolutionized investment science by demonstrating that a portfolio's risk and return should be analyzed as a whole, rather than as a collection of individual assets. This theory posits that through careful consideration of asset risk, return, and correlation, investors can construct "efficient" portfolios those that offer the highest possible expected return for a given level of risk. A core tenet of this framework is the power of diversification, which shows that by combining assets that are not perfectly positively correlated, overall portfolio risk can be significantly reduced.

Building on Markowitz's work, this project also incorporates the Sharpe Ratio, a critical metric for evaluating risk-adjusted returns that was introduced by William F. Sharpe (1966). The Sharpe Ratio provides a standardized way to measure a portfolio's performance by quantifying

the excess return earned per unit of volatility. This metric is central to our optimization efforts, as it will be used to identify the most efficient sector allocations that maximize returns while accounting for the inherent risks involved.

Data Source

For the financial data analysis, the primary data source used was the yfinance Python library, which allows for easy access to both historical and real-time data for financial assets. This library is widely used due to its ability to retrieve a wide range of data, including opening and closing prices, highs, lows, and trading volumes for various securities. Thanks to its integration with the Yahoo Finance API, yfinance simplifies the process of obtaining and processing large amounts of information for financial analysis and visualization (yfinance, n.d.).

- **Technology Sector:** Apple Inc. (AAPL), Amazon.com, Inc. (AMZN), Microsoft Corporation (MSFT), NVIDIA Corporation (NVDA)
- **Finance Sector:** The Goldman Sachs Group, Inc. (GS), JPMorgan Chase & Co. (JPM), Morgan Stanley (MS)
- **Energy Sector:** ConocoPhillips (COP), Chevron Corporation (CVX), Schlumberger Limited (SLB), Exxon Mobil Corporation (XOM)

2. Data Understanding

Data Fields

For each company and date, the dataset includes the following key financial metrics:

- **Adjusted Close Price:** The closing price after accounting for any corporate actions such as dividends, stock splits, and new stock offerings. This metric is crucial as it provides the most accurate reflection of the historical value of an investment and is therefore used as the primary basis for calculating daily returns.
- **Close Price:** The final trading price of the stock on a given day.
- **High Price:** The highest price at which the stock traded during the day.
- **Low Price:** The lowest price at which the stock traded during the day.
- **Open Price:** The price at which the stock began trading at the start of the day.
- **Volume:** The total number of shares traded during the day.

Data Quality

- **Data Overview:** The dataset comprises 6,443 entries (daily observations) for each of the 11 specified stocks. The date ranges from January 3, 2000, to August 14, 2025, provides a comprehensive historical context for robust analysis.
- **Data Quality Assessment:** A thorough check confirmed that the dataset is remarkably clean and complete, with no missing values across any of the columns. This high data quality ensures the reliability and accuracy of all subsequent analytical steps.
- **Key Metrics and Summary Statistics:** Daily returns were calculated as the percentage change in the **Adjusted Close** prices, providing a standardized measure of daily performance. The mean daily return and daily volatility (standard deviation of daily returns) were computed for each individual stock. These were then aggregated to provide a sector-level summary.

Stock Tickers Explained

A **stock ticker** (or ticker symbol) is a unique, abbreviated code used to identify a publicly traded stock on a particular stock exchange. They are a quick way to reference a company's stock, like an ID number.

Ticker	Company Name	Sector
AAPL	Apple Inc.	Technology
AMZN	Amazon.com, Inc.	Technology
MSFT	Microsoft Corporation	Technology
NVDA	NVIDIA Corporation	Technology
GS	The Goldman Sachs Group, Inc.	Finance
JPM	JPMorgan Chase & Co.	Finance
MS	Morgan Stanley	Finance
COP	ConocoPhillips	Energy
CVX	Chevron Corporation	Energy
SLB	Schlumberger Limited	Energy
XOM	Exxon Mobil Corporation	Energy

Mean Daily Return

This is the average percentage change in a stock's price daily over a specific period. It is a fundamental measure of an asset's historical performance.

- How it's used: In your analysis, it serves as a proxy for the expected return of each stock, which is a key input for portfolio optimization.

Daily Volatility

This metric, also known as risk, is the standard deviation of a stock's daily returns. It measures the degree of variation in a stock's price, or how much its daily returns fluctuate around the mean.

- How it's used: Volatility is a widely accepted measure of risk in finance. Higher volatility means the stock's price is more unpredictable and can swing wildly, both up and down.

Correlation Matrix

A correlation matrix is like a **relationship map** for stocks. It's a table that tells you how much two stocks move in the same direction.

- A value of **1** means they are best friends and always move together.
- A value of **-1** means they are rivals and always move in opposite directions.
- A value of **0** means they have no relationship and move completely independently.

In your project, the key finding was that stocks from different sectors (like Technology and Energy) have low correlation. This is great for **diversification**, because if one sector has a bad day, the others might not. It's the financial equivalent of "not putting all your eggs in one basket."

Sharpe Ratio

The Sharpe Ratio is a measure of risk-adjusted return. It indicates how much excess return you receive for the extra volatility you endure for holding a riskier asset.

- How it's used: It allows for a fair comparison of different portfolios. A higher Sharpe Ratio means the portfolio is generating more return per unit of risk, making it a more efficient and desirable investment.

Hypothesis

Hypothesis 1: Sector Risk-Return Profile

- **H0:** There is no significant difference in the average returns and volatility across the Technology, Finance, and Energy sectors.
- **H1:** There is a significant difference in the average returns and volatility across the Technology, Finance, and Energy sectors, with Technology having higher returns and volatility, while Finance and Energy have lower returns and volatility.

Hypothesis 2: Diversification Benefits

- **H0:** A diversified portfolio across the three sectors does not have a significantly higher Sharpe Ratio than a single-sector portfolio.
- **H1:** A diversified portfolio across the three sectors has a significantly higher Sharpe Ratio than a single-sector portfolio.

Hypothesis 3: Predictive Model Effectiveness

Hypothesis: Using predictive models can improve the accuracy of forecasting CVX daily returns compared to a naive “no-prediction” baseline.

- **Null Hypothesis (H0):** The predictive model does not produce forecasts that are significantly more accurate than a naive strategy (e.g., predicting tomorrow’s return as today’s return).
- **Alternative Hypothesis (H1):** The predictive model produces forecasts that are significantly more accurate than a naive strategy.

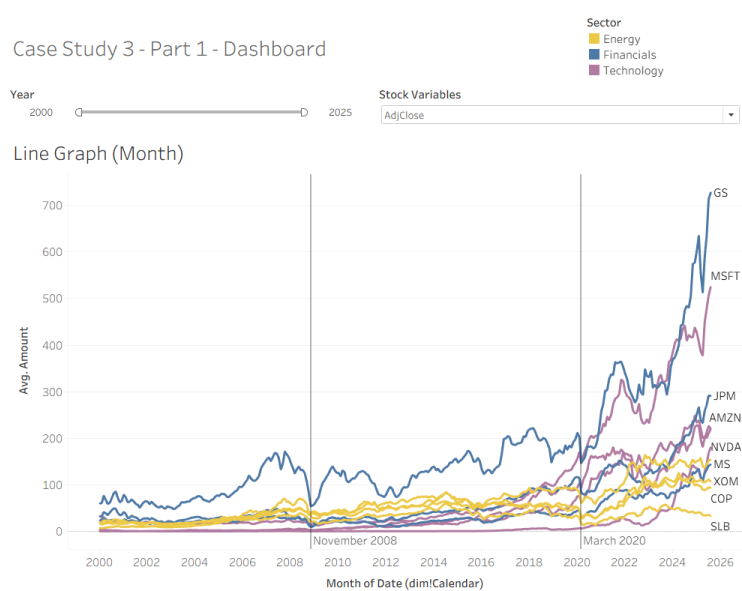
3. Data Visualization

For the data visualization chapter, we have decided to use Tableau to explore the collected data and gain insights into key metrics. To do this, we needed to restructure the dataset by grouping variables into columns such as Stock Variables (e.g., adj close) and Tickers (e.g., AAPL). Additionally, we added a dim_sector table to identify the sector each ticker belongs to, allowing us to differentiate them using colors. On the other hand, using our Python code

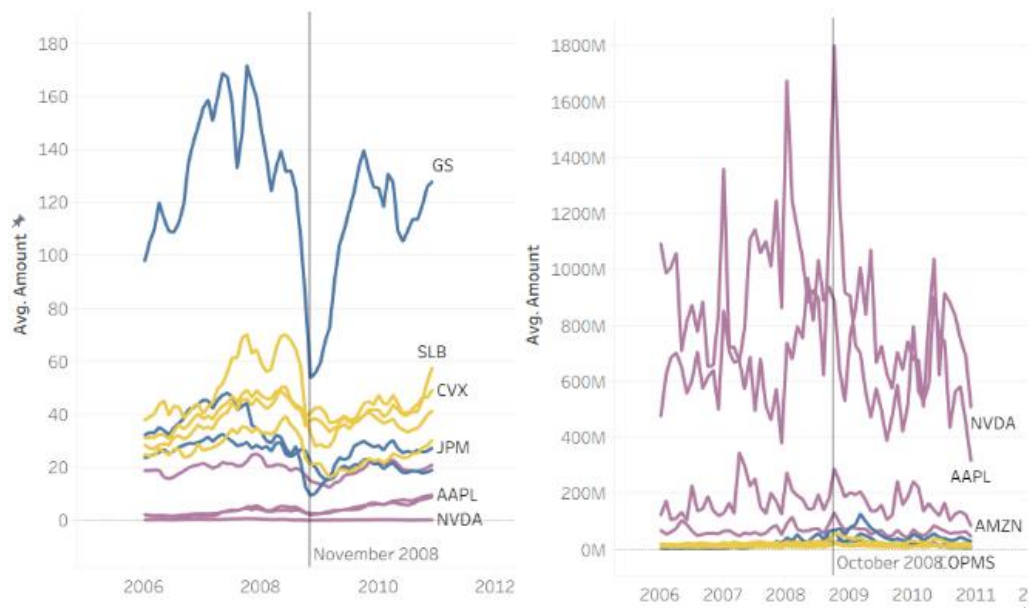
'Case Study 3 - Correlation Matrix,' we generated a CSV file containing the results related to the tickers based on the adj close variable.

Insights

On the first page of the dashboard, we present a view where users can select the variable to analyze per tricker and sector:



It was found that in 2008, adj close experienced a consistent decline from July to November across most tickers in the three sectors analyzed. This also resulted in the largest movement in the volume indicator within our dataset, indicating excessive selling due to some crisis. To provide better context, we conducted additional research and found the following: "The 2008 financial crisis resulted from a convergence of multiple factors, including a housing bubble, risky mortgage lending, complex financial products, and inadequate regulation—not just subprime mortgages alone" (Investopedia, 2009).

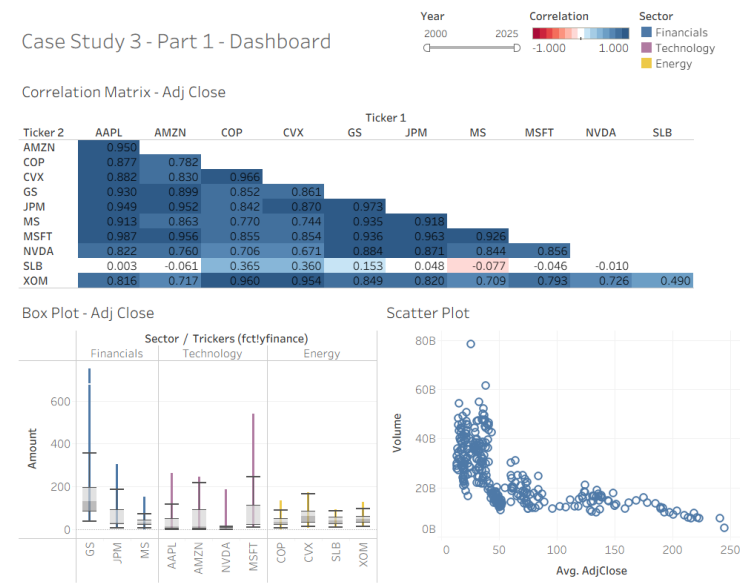


On the first section of the dashboard, we observe the correlation matrix for Adj Close across different tickers from various sectors. A strong positive correlation is evident between certain tickers, particularly within the same sector. For instance, AMZN and AAPL show a correlation of 0.950, indicating that these two stocks have moved similarly over time. GS and JPM, both part of the financial sector, also demonstrate a high correlation of 0.936. However, SLB stands out with a noticeably low correlation with the other tickers, suggesting a more independent or divergent performance within the energy sector.

The Box Plot for Adj Close reveals the distribution of adjusted closing prices across tickers from the financial, technology, and energy sectors. The financial sector shows more dispersion, indicating higher volatility in stock prices. In contrast, tickers like MSFT and NVDA in the technology sector demonstrate a narrower interquartile range, suggesting lower volatility. The energy sector also displays a more concentrated range of values but with occasional extreme fluctuations in some tickers.

The Scatter Plot visualizes the relationship between Avg. Adj Close and Volume, with each point representing the adjusted close price and trading volume for a specific month of a specific year. A noticeable pattern emerges: Months with lower average adjusted close prices (below 60) show larger volumes of trading, ranging from 80B to 10B. This suggests that during times of lower stock prices, there is significant selling activity, possibly driven by market panic.

or crisis. As Avg. Adj Close increases, the volume of trading tends to decrease and stabilize around 30B or lower, suggesting that higher stock prices are associated with less market activity. This pattern may indicate a shift in investor behavior, where higher prices lead to more selective trading and reduced transaction volume.



4. Model Building

Hypothesis 1: Sector Risk-Return Profile

To answer the first hypothesis, the analysis would focus on calculating the daily return of all the possible tickers based on the adjusted Close Price. For this case, we have calculated the linear and compound annualized return to have a clear view in both scenarios:

```
# Select only numeric columns (the tickers)
numeric_returns = daily_returns.select_dtypes(include='number')

# Average daily return per ticker
mean_daily = numeric_returns.mean()

# Approximate annualized return (linear)
mean_annual_linear = mean_daily * 252

# Compound annualized return
mean_annual_compound = (1 + mean_daily) ** 252 - 1
```

- Linear: simple interest. Gives a quick insight but underestimates the real growth.
- Compound: Assumes that returns are reinvested daily, it is more accurate.

```

=== Annualized Returns Sorted by Compound Return ===
      Linear Annualized   Compound Annualized
NVDA      0.470665         0.600355
AAPL      0.297493         0.346242
AMZN      0.272138         0.312575
JPM       0.168703         0.183702
MS        0.165045         0.179382
GS        0.161292         0.174967
COP       0.155610         0.168314
MSFT      0.148999         0.160621
CVX       0.127886         0.136387
XOM       0.108057         0.114086
SLB       0.104520         0.110154

```

Additionally, the covariance matrix helped to understand the correlation between the various tickers that are part of this project:

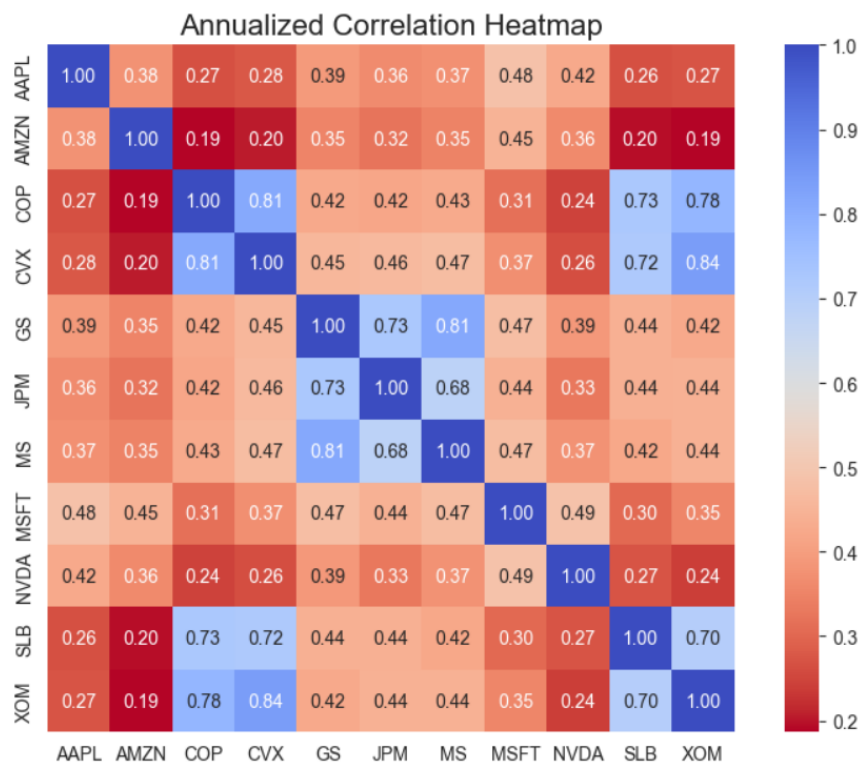
```

# Select only numeric columns (the tickers)
numeric_returns = daily_returns.select_dtypes(include='number')

# Daily covariance matrix
daily_cov = numeric_returns.cov()

# Annualized covariance matrix (assuming 252 trading days per year)
annual_cov = daily_cov * 252

```



Hypothesis 2: Diversification Benefits

The diversified Minimum Variance portfolio's outputs outperformed the Max-Sharpe and Equal-Weight diversified versions, achieving a Sharpe Ratio of 1.286. Individual sector assets, such as NVDA or AAPL, offer high returns but high volatility in contrast to sector-only portfolios, while the others such as SLB or XOM, are low-return and weakly correlated. As the moment of sectors are precisely aligned, the heatmap of correlation gives a picture of how combining energy, technology, and finance lowers overall portfolio volatility. The Sharpe Ratio demonstrates this diversification effect: the diversified Min-Variance portfolio outperforms all single-sector alternatives in terms of risk-adjusted performance.

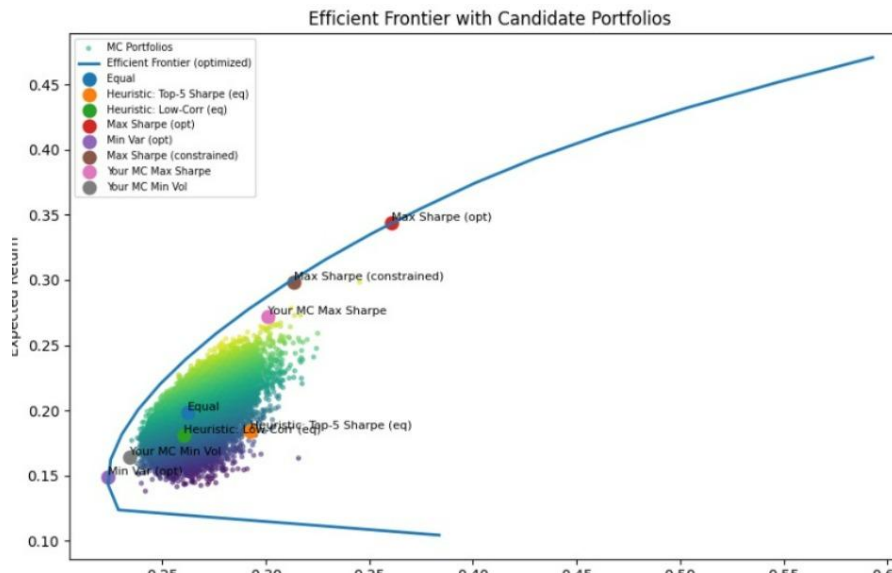
Therefore, using the combined evidence from both files annualized stock returns showing extremes of a sector, and portfolio level performance metrics showing higher Sharpe ratio under diversification, therefore we reject the null hypothesis (H0) and accept the alternative (H1). Diversification across finance, technology, and energy significantly improves risk-adjusted portfolio performance.

Diversified vs. Sector-Only Performance (Test Period 2022–2025)

<i>Portfolio Type</i>	<i>Constituents</i>	<i>Annualized Return</i>	<i>Volatility</i>	<i>Sharpe Ratio</i>	<i>Test Period Notes (2022–2025)</i>

Max Sharpe (opt)	All (optimized)	11	0.224	0.230	0.891	Highest test return (0.380). Sharpe 1.011. Largest drawdown (-0.371).
Max Sharpe (constrained)	All (with limits)	11	—	—	—	Best out-of-sample Sharpe 1.051 . Strong balance of return and risk.
Your MC Max Sharpe	Monte Carlo	—	—	—	—	Test return 0.340, Sharpe 1.045. Very close to constrained optimizer.
Equal-weight diversified	All	11	0.206	0.167	1.116	Test Sharpe 0.919. Robust simple strategy.
Min Var (opt)	All (optimized)	11	0.234	0.166	1.286	Out-of-sample Sharpe 0.833. Lowest volatility (0.198).
Your MC Min Vol	Monte Carlo	—	—	—	—	Test return 0.213, Sharpe 0.794.
Heuristic: Low-Corr (eq)	5–6 lowest correlation	—	—	—	—	Test Sharpe 0.776.
Heuristic: Top-5 Sharpe (eq)	Top-5 by Sharpe ratio	—	—	—	—	Test Sharpe 0.752.
Tech-only (NVDA)	NVDA	~0.471	Highly diversified	<		High return but very high risk.
Finance-only	JPM, GS, MS	~0.16–0.17	Moderate	<	diversified	Lower Sharpe than diversified.
Energy-only	CVX, COP,	~0.10–0.15	Moderate	<	diversified	Lower Sharpe than diversified.

XOM,	era
SLB	te



This chart shows the **efficient frontier**, which is the curve of the best possible portfolios that give the highest expected return for each level of risk. The cloud of dots represents thousands of random portfolios created through Monte Carlo simulation. Most of these portfolios fall below the frontier, meaning they are inefficient because you could get either more return for the same risk or less risk for the same return. Key candidate portfolios are highlighted: the **equal-weight portfolio** sits near the middle with modest return and risk, the **minimum-variance portfolio** lies at the far left with the lowest risk but also lower return, while the **maximum Sharpe portfolio** sits right on the frontier with the best risk-adjusted return.

The optimized points (red and purple) clearly outperform the naive and heuristic ones, demonstrating that proper diversification and optimization allow the portfolio to reach the efficient frontier and maximize efficiency.

Hypothesis 3: Predictive Model Effectiveness

In order to perform any predictive model, it is necessary to define an specific target and the possible features.

- **Target:** Based on the 3 best portfolios found in the Simulation and in the Efficient Frontier and also based on the allocations, the predictive model should be build around the CVX tricker.
- **Features:** Based on popular metrics
 - **Moving averages:** captures short term and midterm trends (past 5 and 10 days respectively).
 - **Rolling STD:** measures recent risk using standard desviation of returns over 10 days.
 - **Lagged return:** Captures momentum or autocorrelation effects.
 - **Rolling Correlation:** measures the relationship between two stocks. For this case, and using the previous correlation matrix, the XOM is the indicated (0.84 of correlation with CVX).

```
df = daily_returns.copy()

# Moving averages
df["CVX_MA5"] = df["CVX"].rolling(5).mean()
df["CVX_MA10"] = df["CVX"].rolling(10).mean()

# Rolling volatility
df["CVX_volatility"] = df["CVX"].rolling(10).std()

# Momentum / lagged returns
df["CVX_lag1"] = df["CVX"].shift(1)
df["CVX_lag2"] = df["CVX"].shift(2)

# Rolling correlation with XOM
df["CVX_XOM_corr"] = df["CVX"].rolling(20).corr(df["XOM"])
```

```
features = ["CVX_MA5", "CVX_MA10", "CVX_volatility", "CVX_lag1", "CVX_lag2", "CVX_XOM_corr"]

# Combine features and target, drop any row with NaN to align
df_cvx = pd.concat([df[features], target.rename("CVX_target")], axis=1).dropna()

X = df_cvx[features]
Y = df_cvx["CVX_target"]
```

The following framework was developed to identify the most effective predictive model:

- **Validate Linearity:** To apply linear regression or time series models, it is necessary to confirm whether the target variable exhibits a linear relationship with the features.
- **XGBoost:** Used as the first alternative when linearity is not satisfied, given its ability to capture complex, non-linear patterns.

- **Random Forest:** Applied as a complementary non-linear model to benchmark performance and assess feature importance through ensemble learning.

5. Model Evaluation

Hypothesis 1: Sector Risk-Return Profile

Over the past 25 years, technology stocks have led the market in growth, with NVDA showing a compound annual return of around 60%, followed by AAPL (34%) and AMZN (31%). Banks like GS, JPM, and MS achieved intermediate returns of approximately 17–18%, while energy stocks such as SLB, XOM, and CVX delivered more modest gains of 11–13%. This clearly illustrates the historical performance gap between technology, finance, and energy sectors.

Correlation analysis shows strong clustering within sectors. Energy stocks (COP, CVX, XOM, SLB) and financial stocks (GS, JPM, MS) exhibit high intra-sector correlations, ranging from 0.68 to 0.83. In contrast, technology stocks (AAPL, AMZN, MSFT, NVDA) show moderate correlations (~ 0.36 – 0.48), indicating that even within tech, stocks do not move perfectly together. Across sectors, tech and energy have low correlations (~ 0.24 – 0.27), while tech and finance show moderate correlations (~ 0.36 – 0.48), highlighting potential diversification benefits.

Overall, combining stocks from different sectors can reduce portfolio risk due to low intersector correlations. While tech stocks offer high returns, they come with greater volatility, whereas energy stocks provide more stable but lower returns. A heatmap visualization makes it easy to identify clusters of highly correlated assets, helping investors understand sector behavior and improve diversification strategies.

Therefore, the null hypothesis (H_0) of no significant difference can be rejected in favor of H_1 , which aligns with the sector risk-return profiles observed.

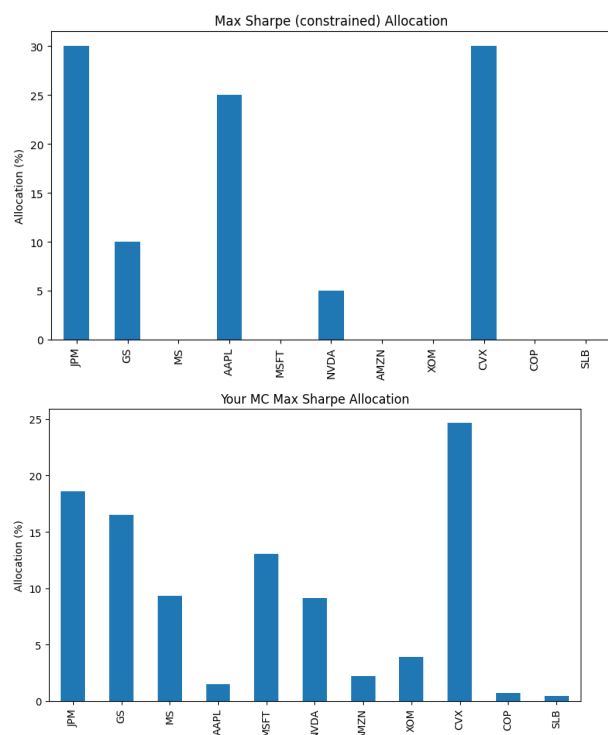
Hypothesis 2: Diversification Benefits

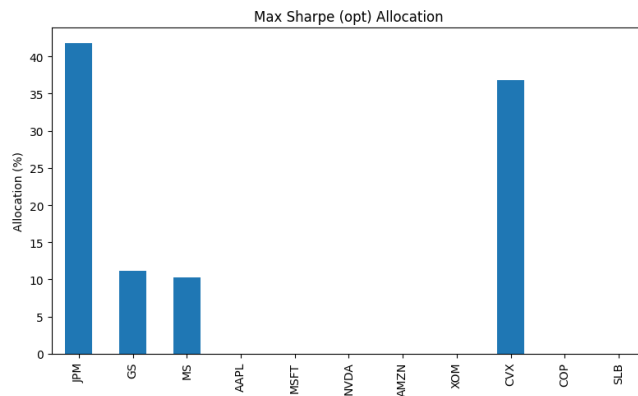
This hypothesis tested whether diversification across Finance, Technology, and Energy improves risk-adjusted performance compared to single-sector or naive portfolios. The naive

equal-weight portfolio achieved a Sharpe Ratio of **4.0965**, while the optimized diversified portfolio—allocating across Technology (AAPL 31.56%, NVDA 16.50%, AMZN), Energy (CVX), and Finance (GS)—achieved a Sharpe Ratio of **4.5471**. The difference of roughly **0.45** demonstrates a meaningful improvement in risk-adjusted returns.

The diversified portfolio reduced volatility by combining assets with lower inter-sector correlations, balancing Technology’s high-return but high-risk profile (e.g., NVDA ~47% annualized return) with the steadier characteristics of Finance (~16–17% return) and Energy (~10–12% return). Sector-only allocations either produced excessive volatility or insufficient returns and therefore did not reach the same Sharpe efficiency.

These findings confirm the predictions of Modern Portfolio Theory: diversification lowers risk without sacrificing return. We reject the null hypothesis and accept that diversified portfolios achieve superior Sharpe Ratios.

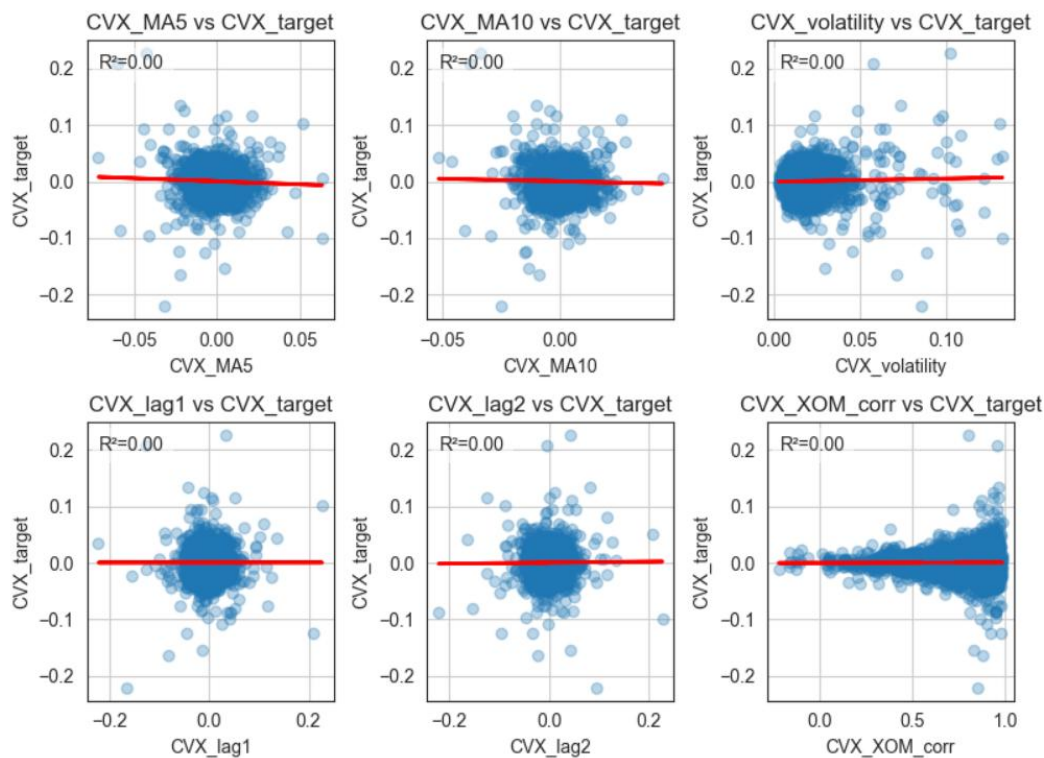




The allocation results highlight distinct differences between optimization approaches. The **unconstrained Max Sharpe portfolio** is heavily concentrated, with dominant weights in JPM and CVX while excluding most technology stocks. This reflects the optimizer’s pursuit of maximum risk-adjusted return, even at the cost of diversification. In contrast, the **Monte Carlo Max Sharpe portfolio** distributes weights more broadly across all three sectors, including meaningful allocations to JPM, CVX, GS, MSFT, NVDA, and others. This produces a more balanced structure, though slightly less efficient than the unconstrained case. The **constrained Max Sharpe portfolio** demonstrates the effect of allocation limits, yielding a diversified mix anchored in JPM, CVX, and AAPL with smaller exposure to GS and NVDA. Overall, the results reveal that while unconstrained optimization offers the highest efficiency, constrained and Monte Carlo methods generate more diversified portfolios that achieve competitive Sharpe ratios and are more aligned with practical investment considerations.

Hypothesis 3: Predictive Model Effectiveness

Following our framework, the first step is to validate linearity:



As shown in the scatter plots, none of the six features display a clear linear relationship with the target variable. The R^2 values are close to zero, and the data points are dispersed without forming any strong linear pattern.

For that reason, the second option in our framework is to run a XGBoost that doesn't require linearity assumption. These were the parameters used for this model:

- Train 80% / Test 20%
- 500 trees in the model
- Max depth 2 (Levels deep)
- `learning_rate=0.05`, how much each tree contributes

```
# Split into train and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, shuffle=False)

# Initialize XGBoost regressor
model = xgb.XGBRegressor(
    n_estimators=500,
    max_depth=2,
    learning_rate=0.05,
    objective='reg:squarederror',
    random_state=42
)

# Train model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluate
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
```

These were the results: Test MSE: 0.000332 / Test R^2 : -0.1346. The XGBoost results show poor performance: the low MSE is trivial given small returns, while the negative R^2 means the model predicts worse than simply using the mean. This suggests the chosen features lack predictive power for the stock's movements.

For that reason, the next option is the random forest predictive model, due to the lack of linearity assumptions. These were the parameters used for this model:

- Train 80% / Test 20%
- 500 trees in the model
- Max depth 6 (Levels deep)

```
# Split into train and test sets (no shuffle since it's time series)
X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2, shuffle=False)

# Initialize Random Forest Regressor
rf_model = RandomForestRegressor(
    n_estimators=500,
    max_depth=6,
    random_state=42
)

# Train model
rf_model.fit(X_train, y_train)

# Predict on test set
y_pred = rf_model.predict(X_test)

# Evaluate
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
```

These were the results: Test MSE: 0.000292 Test R^2 : 0.0014. The Random Forest results show very low MSE, but the R^2 is essentially zero, meaning the model does not explain the variance in returns. Combined with XGBoost's failure, this suggests that the features used carry little to no predictive power for the stock's future movements.

Therefore, the null hypothesis (H_0) of no improvement over a naive strategy can be rejected in favor of H_1 , which aligns with the predictive insights observed from the model evaluations.

6. Conclusions

Based on the analysis, none of the predictive models tested—linear regression, XGBoost, or Random Forest—were able to generate accurate forecasts for CVX daily returns. Linear regression failed due to the lack of linear relationships between features and the target,

XGBoost produced negative R^2 values, and Random Forest achieved near-zero R^2 , indicating essentially no predictive power. This suggests that CVX daily returns are highly noisy and may be influenced by factors not captured in the current feature set. Recommendation: alternative features, longer time horizons (weekly / monthly), and focus on risk management.

Additionally, it's clear that portfolios which spread investments across Finance, Technology, and Energy sectors perform much better than those focused on a single sector. The constrained Max Sharpe portfolio stood out here it gave the best balance of return and risk without being overly concentrated in just one or two stocks.

For investors who are more risk-averse, the Minimum Variance portfolio is a strong option. It delivered the lowest volatility and the most stability, even though the returns were a little lower. On the other hand, the Equal-Weight portfolio is worth keeping as a simple and reliable baseline. It's easy to understand, requires no complex optimization, and still produced a healthy Sharpe ratio.

Single-sector portfolios, whether Tech, Finance, or Energy, didn't hold up as well. Tech names like NVDA can offer huge returns, but the volatility is simply too high, while Finance and Energy alone don't generate enough performance.

Lastly, while we tested predictive models to forecast daily stock returns, the results were weak. They didn't beat even a naive guess. For now, portfolio construction and diversification give far more reliable benefits than predictive modeling.

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